

PART 1 →

$$y = x^2 + 5 \quad (0,0) \quad (-4,0) \quad (-8,0) \quad (2,0) \quad (6,0)$$

analytical

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2$$

$$D(x) = (x - x_0)^2 + [(x^2 + 5) - y_0]^2$$

$$D'(x) = 2(x - x_0) + 2[(x^2 + 5) - y_0](2x)$$

$\downarrow$ simplify $y_0 = 0$
 (for all given pts)

$$0 = 2x - 2x_0 + 4x^3 + 20x$$

$$0 = 2x^3 + 11x - x_0$$

→ for each point, solve equation for x

$$① (0,0): 2x^3 + 11x = 0 \Rightarrow x(2x^2 + 11) = 0$$

$$x = 0$$

$$\hookrightarrow f(0) = 0^2 + 5 = 5$$

$$d = \sqrt{(0-0)^2 + (5-0)^2}$$

$$d = 5$$

$$② (-4,0): 2x^3 + 11x + 4 = 0 \Rightarrow x = -0.3555$$

$$\hookrightarrow f(-0.3555) = 5.1264$$

$$d = \sqrt{(-0.3555 - (-4))^2 + (5.1264 - 0)^2}$$

$$d = 6.28$$

$$③ (-8,0): 2x^3 + 11x + 8 = 0 \Rightarrow x = -0.6721$$

$$\hookrightarrow f(-0.6721) = 5.4517$$

$$d = \sqrt{(-0.6721 - (-8))^2 + (5.4517 - 0)^2}$$

$$d = 9.13$$

$$④ (2,0): 2x^3 + 11x - 2 = 0 \Rightarrow x = 0.1807$$

$$\hookrightarrow f(0.1807) = 5.0827$$

$$d = \sqrt{(0.1807 - 2)^2 + (5.0827 - 0)^2}$$

$$d = 5.35$$

$$⑤ (6,0): 2x^3 + 11x - 6 = 0 \Rightarrow x = 0.5199$$

$$\hookrightarrow f(0.5199) = 5.2703$$

$$d = \sqrt{(0.5199 - 6)^2 + (5.2703 - 0)^2}$$

$$d = 7.60$$

* only $x=0$ is exact, other cubic roots found with graphical inference.

newton-raphson

(same steps followed for ALL points, only $(-4,0)$ shown below)set up distance formula (dropped $\sqrt{}$): $D(x) = (x - x_0)^2 + (f(x) - y_0)^2$

$$\text{for point } (-4,0) \rightarrow D(x) = (x + 4)^2 + (x^2 + 5)^2 = x^4 + 11x^2 + 8x + 41$$

$$\rightarrow D'(x) = 4x^3 + 22x + 8$$

$$D''(x) = 12x^2 + 22$$

$$\text{iteration rule: } x_{n+1} = x_n - \frac{D'(x_n)}{D''(x_n)}, \text{ initial guess: } x_0 = 0$$

$$\text{iteration 1: } D'(0) = 4(0)^3 + 22(0) + 8 = 8, \quad D''(0) = 12(0)^2 + 22 = 22$$

$$x_1 = 0 - \frac{8}{22} = -0.3636$$

$$\text{iteration 2: } x_2 = -0.3636 - \frac{(-0.19148)}{(23.5865)} = -0.35548$$

$$\text{iteration 3: } x_3 = -0.35548 - \frac{(-0.00024239)}{(23.51639)} = -0.355469$$

$$\text{iteration 4: } x_4 = -0.355469 - \frac{(-0.00000542)}{(23.51630)} = -0.35547$$

} difference $< e^{-7}$ tolerance ✓

$$\Rightarrow x^* = -0.35547$$

$$D(x^*) = (-0.35547 + 4)^2 + (0.35547^2 + 5)^2 = 39.5622$$

$$\Rightarrow d = \sqrt{39.5622} \approx 6.29$$

golden section search (same steps followed for all points, only few steps shown below for $(-4, 0)$)

$$\text{SETUP} \left\{ \begin{array}{l} D(x) = (x+4)^2 + (x^2+5)^2 ; \text{ first iteration interval: } [-2, 2] \\ \psi = 1.618, \text{ resphi} = 0.38196 \\ [a, b] \end{array} \right.$$

$$\text{iteration 1: } x_1 = a + \text{resphi}(b-a) = -2 + (0.38196)(2 - (-2)) = -0.472136$$

$$x_2 = b - \text{resphi}(b-a) = 2 - (0.38196)(4) = 0.472136$$

$$D(x_1) = (-0.472136 + 4)^2 + ((-0.472136)^2 + 5)^2 = 39.72464$$

$$D(x_2) = (0.472136 + 4)^2 + (0.472136^2 + 5)^2 = 47.27882$$

$\left. \begin{array}{l} D(x_1) < D(x_2) \therefore \text{min. lies in } [a, x_2] \\ \text{new range for iteration 2} \end{array} \right\}$

:

* continue with iterations till convergence ; results in same output as prev. 2 methods

$$x^* = -0.35547, d \approx 6.28$$